Midterm 1 review

Exam format

Ten short essay questions (100-250 words should be enough to answer each).

Three or four gradeable items each. (Lots of partial credit available.)

Open book. (No need to memorize names and dates.)

Theme 1: Phonemes

Note this chart is organized by *features:*

PLACE (x-axis: front to back of vocal tract)
MANNER (y-axis: degree of closure – closed to open)
VOICING (pairs: onset of voicing before/during or after closure)

CONSONANTS (PULMONIC) © 2005 IPA

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k g	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ⱱ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Where symbols appear in pairs, the one to the right represents a voiced consonant. Shaded areas denote articulations judged impossible.

/p/

\+

plosive

\+

bilabial

- voiced

/b/

\+

plosive

\+

bilabial

+ voiced

This allows us to describe the similarities and differences between sounds in a relatively precise and well-defined way:

/p/ and /b/ both have the same **manner** and **place of articulation**.

But /p/ and /b/ differ in the timing of phonation relative to the release of the closure **(voicing)**.

This difference is analogous to the difference between /t/ and /d/.

So we can talk in general about “the voiced stops” /b/, /d/, /g/, and so on, which is helpful for phonological descriptions.

Minimal Pairs

Consider how we established that I really do have two different low back vowels.

We had to find two different **words** in which the distinction between them depended on the difference between those vowels.

Minimal pairs are important for demonstrating **contrast**.

If two phones are contrastive -- i.e., you can find a minimal pair where the difference between them results in a difference in meaning -- they are examples of different **phonemes**.



Minimal pairs

Evidence that phonemic contrasts depend on a particular *feature* also comes from minimal pairs.

For example, the pairs /b/ - /p/, /t/ - /d/ and /k/ - /g/ all differ on *voicing* in English. We know this from comparison of **minimal pairs** in each language.

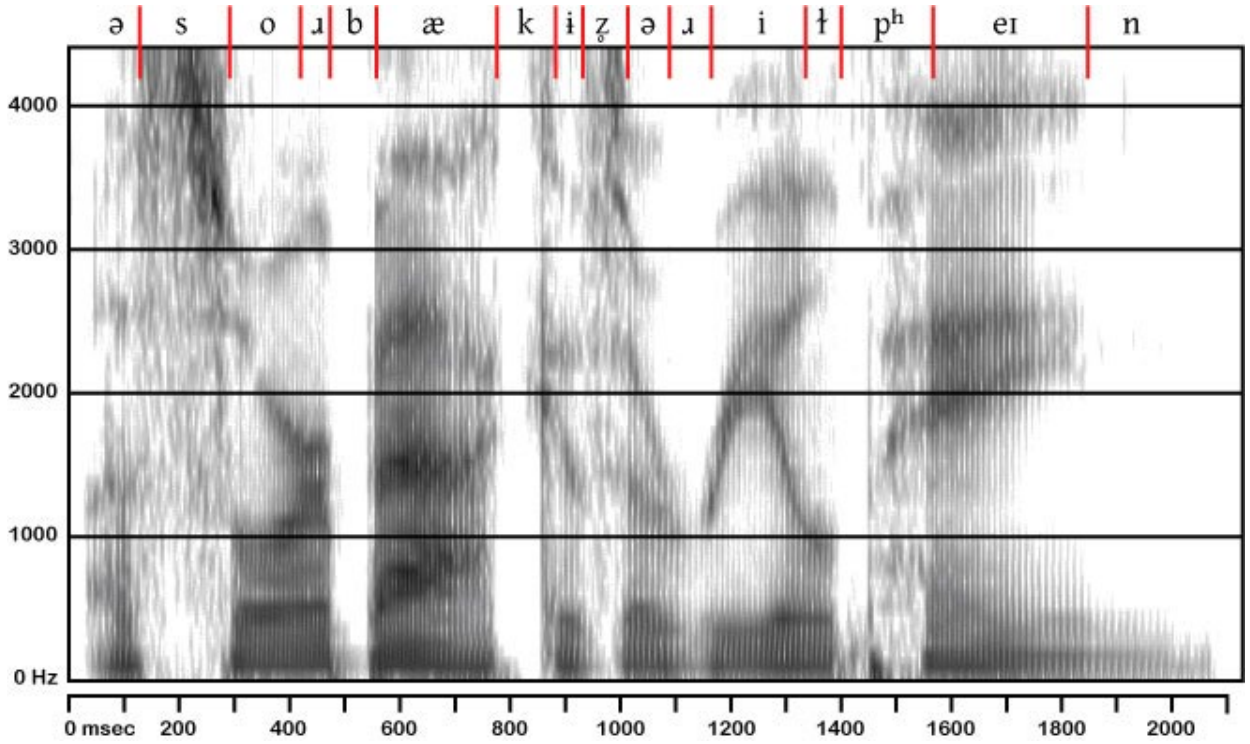
http://www.fon.hum.uva.nl/praat/

Why do I keep insisting that *phonemes are not letters*?

It's easy for literate people to imagine that speech is produced as an orderly stream of letter-like units.

But it is nothing like that.

Here's a spectrogram of the sentence: "A sore back is a real pain."



“The best analogy I have been able to think of for this is a very homely one. Imagine a row of Easter eggs carried along a moving belt; the eggs are of various sizes, and variously colored, but not boiled. At a certain point, the belt carries the row of eggs between the two rollers of a wringer, which quite effectively smash them and rub them more or less into each other. The flow of eggs before the wringer represents the series of impulses from the phoneme source; the mess that emerges from the wringer represents the output of the speech transmitter. At a subsequent point, we have an inspector whose task it is to examine the passing mess and decide, on the basis of the broken and unbroken yolks, the variously spread-out albumen, and the variously colored bits of shell, the nature of the flow of eggs which previously arrived at the wringer. Note that he does not have to try to put the eggs together again—a manifest physical impossibility—but only to identify.

The inspector represents the hearer.”

Hockett (1955)

Theme 2: Morphemes

Derivational Morphology is a process that *derives* new words from existing "parts."

The "parts" (or units) are called **morphemes**.

The word Morphology has a pretty transparent derivation, actually:

morph- (derived from the Greek? for "form")

+

-ology (also from Greek, here meaning something like "the study of")

"the study of the forms of words."

Then a similar process gives you morpheme. We just learned about phonemes. "Phone" is another morpheme from the Greek (for "speech?").

What do we call a unit in text?

What is "graphology?"...oops.

Derivational morphology is productive, but in limited ways

…also with respect to how widely and how regularly affixes are applied.

Consider:

stem + -hood -> a state of being, associated with the stem noun

brotherhood
sisterhood
knighthood (actually kind of odd, because two brothers share their brotherhood, but each knight has his own knighthood..)

*neighborhood -- perfectly good word, but -hood does a different job entirely

*friendhood -- we say friend*ship* instead, for, like, no apparent reason

*enemyhood -- maybe *ought* to be a word. I can only think of top-shelf items like "enmity" or "antagonism" for this concept…

(some examples from Liberman & Prince's introductory Linguistics class at Penn: http://www.ling.upenn.edu/courses/Fall_1998/ling001/morphology2.html Liberman also blogs at http://languagelog.ldc.upenn.edu)

Even **zero derivation** does complicated things with semantics.

Consider these strong associates of the noun “knife.” Are their meanings similar as verbs?

She can **dish** it out but she can’t take it.
He was born with a silver spoon and a **forked** tongue in his mouth.
They decided to **table** the issue after a day of inconclusive negotiations.

If you got really distracted by “silver spoon,” above, you may enjoy reading about Cognitive Linguistics, which we won’t talk about much in this class.

More commonly, derivational morphology changes the pronunciation (and spelling) of a word, e.g., by adding **affixes:**

certificate /sər.ˈtɪ.fɨ.kɨt/
certification /ˌsər.tɨ.fɨ.ˈkeɪ.ʃən/
recertification /ˈri.sər.tɨ.fɨ.ˌkeɪ.ʃən/

Note how adding an affix can impact the phonology of the rest of the word. (This is one reason for “The Chaos” of English spelling, and why higher fidelity to the spoken form doesn’t necessarily improve readability.)

Also consider the noun and verb forms of:

record
invite (“He did not receive an invite”/”We did not invite him”)
entrance (not actually related to each other)

…where the major change is also stress.

Inflectional morphology changes the meaning of words in a more predictable way:

PLURALS
knife(n) - **knives**
one fork - two **forks**
one mouse - a lot of **mice**

3RD PERSON SINGULAR PRESENT
knife(v) - **knifes**
I walk - she **walks**
you run - he **runs**

PAST
knife(v) - **knifed**
Today I walk - yesterday I **walked**
Today she sings - in the past, she **sang**

Menn puts this succinctly:

"...we'd think it was silly to have separate dictionary entries for the singular and plural forms of nouns."

But note that the phonological process of inflecting words often has some **irregularity** to it.

(This will be a big deal later when we talk about different language learning models, and kinds of developmental language impairment.)

Inflectional morphology also depends on the syntax of the rest of the sentence.

PLURALS
knife(n) - **knives**
one fork - two **forks**

…etc.

In the phrase

“two [noun]+pl”

whatever noun goes there *must be* pluralized somehow.

There are no syntactic processes like this for even very productive derivational forms.

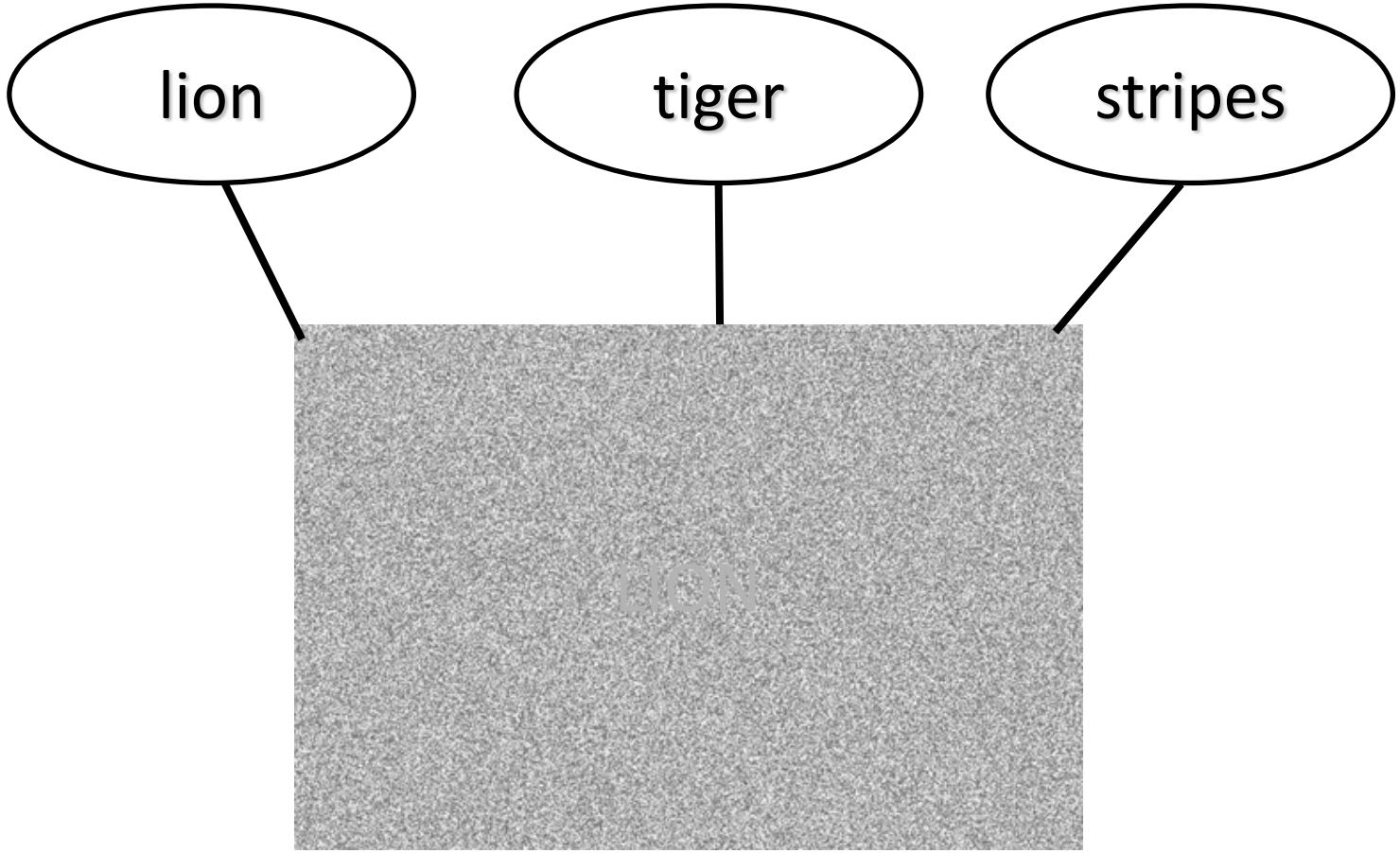
The closest I can think of is a *formula*, like:

“Once you [verb] it, you can un-[verb] it.”

But that’s not exactly syntax.

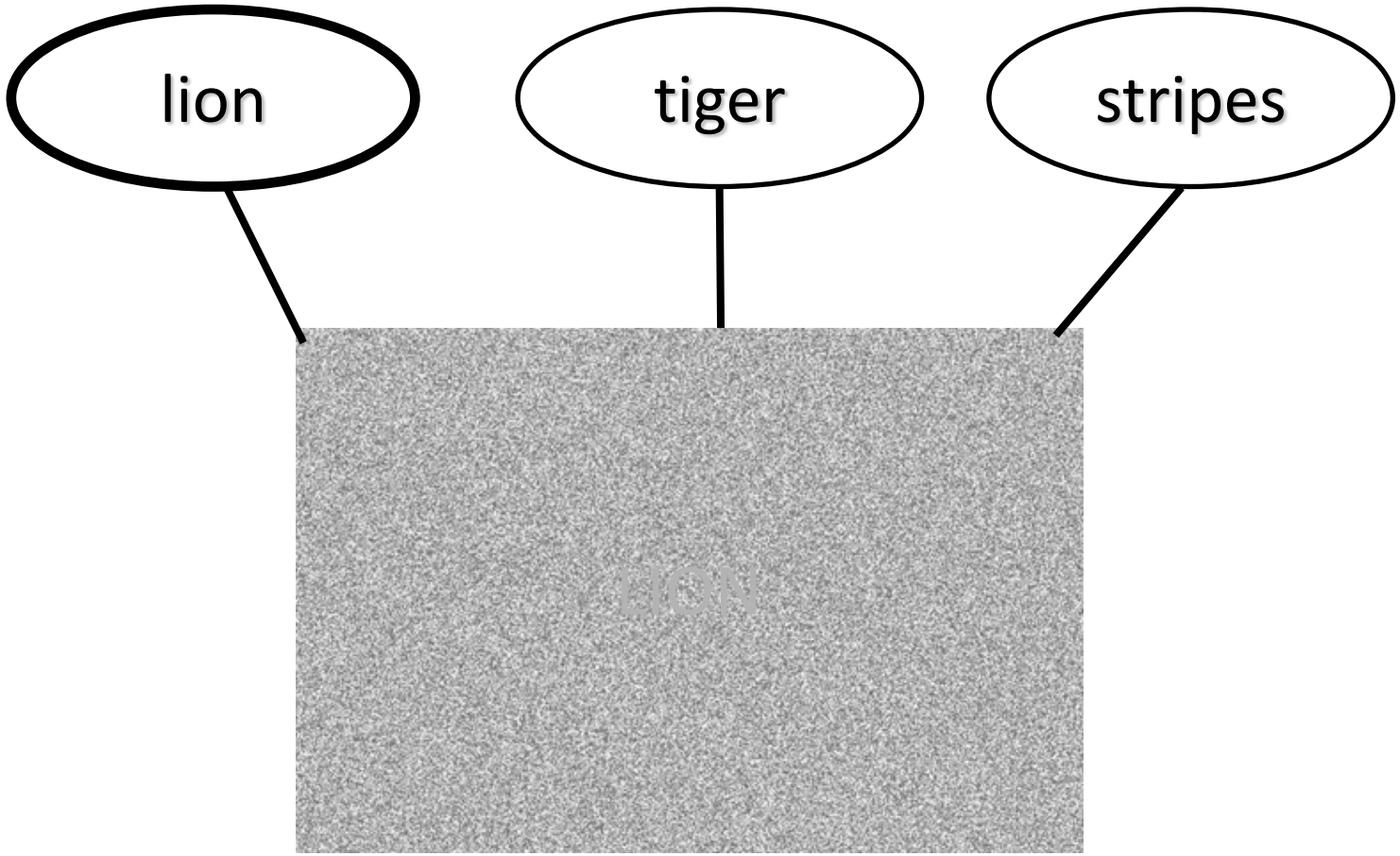
(This will be important when we talk about models of speech production in the next few weeks.)

Theme 3: Activation and its spread



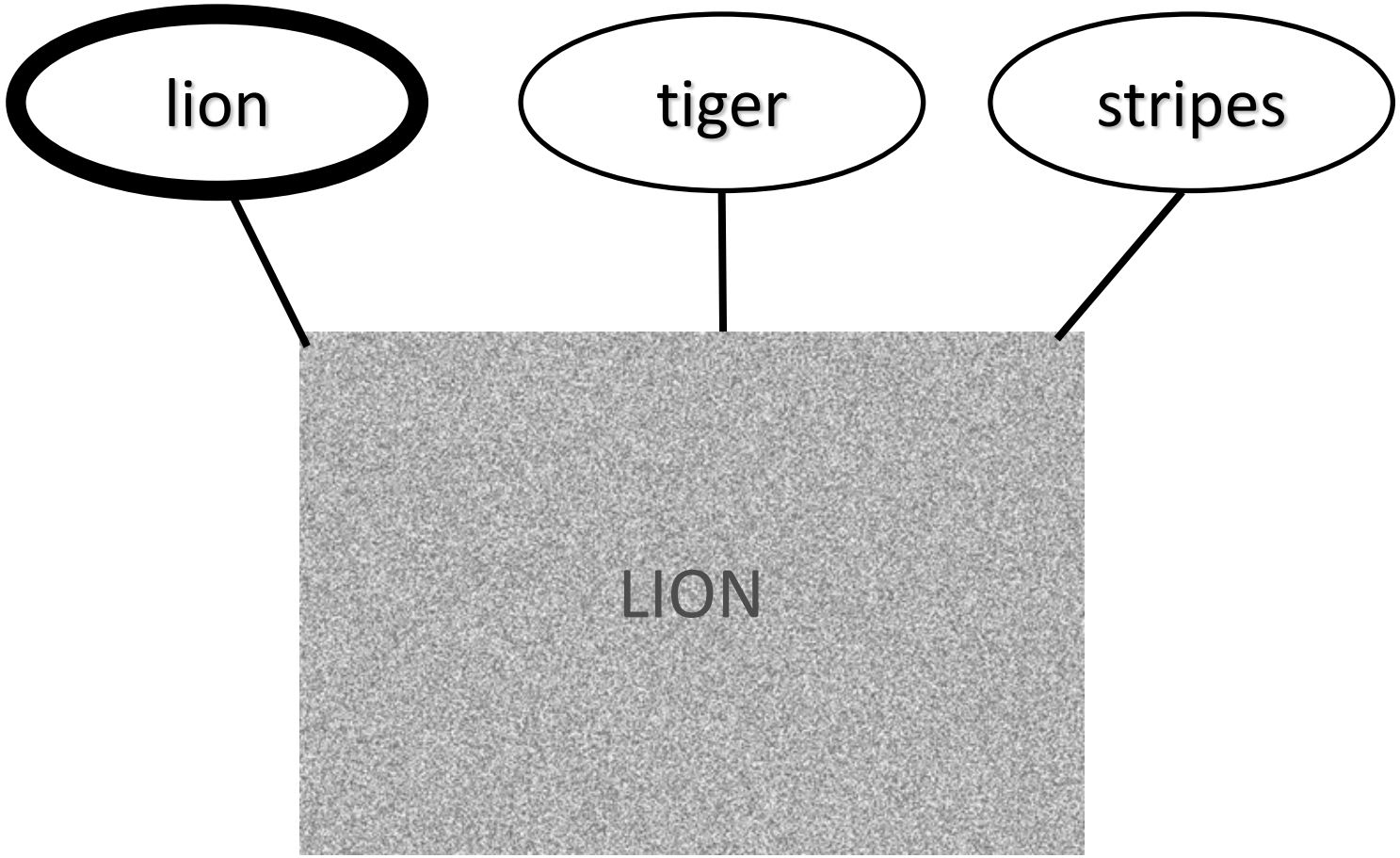
Let's imagine an incredibly simple set of **lexical representations**.

These would be **nodes** in a **network** that can be **activated** by a specific word.



Activation can be partial.

Here, the word LION contrasts poorly with the noise mask, so we might imagine only weak activation as a result.



At some point, the signal-to-noise ratio (SNR) is very favorable.

Activity is strong enough that we can say this simple network has “identified” the word “lion.”

In fact this is an illustration of the “perceptual identification task,” in which participants are asked to push a button as soon as they can identify the item.

Activation dynamics

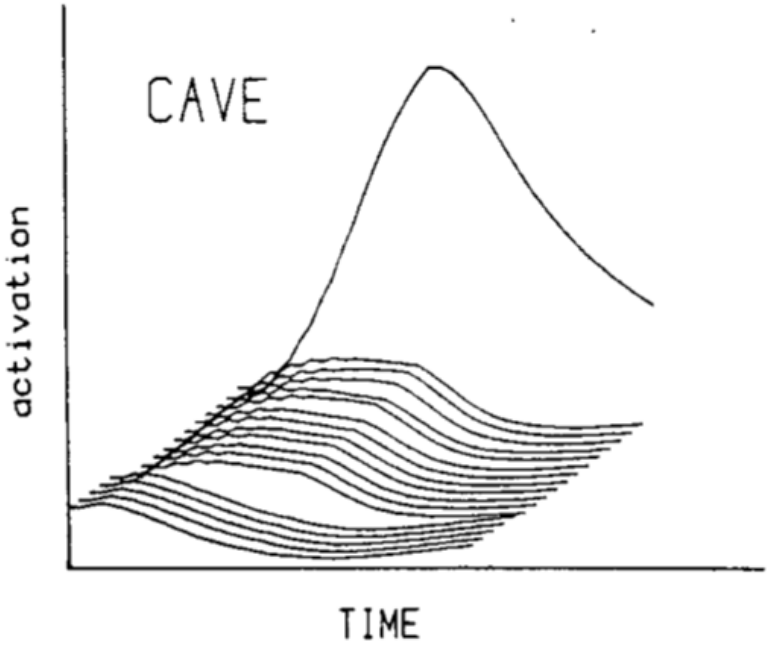


Figure 13. Activity at the word level upon presentation of *CAVE*, with weak letter-to-word inhibition.

Different computational models (like Rumelhart & McClelland's 1986 Interactive Activation model shown here) simulate frequency effects in different ways

Most models of this kind assume that activation unfolds over time.

One way to explain **frequency effects** is to assume that **activation dynamics** are tuned up by experience.

Every time you encounter a word, you might:

raise its baseline level of activation

strengthen incoming connections so it can be activated more quickly

change a threshold parameter...etc.

This is a very simple form of **learning**, of course.

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Activation dynamics

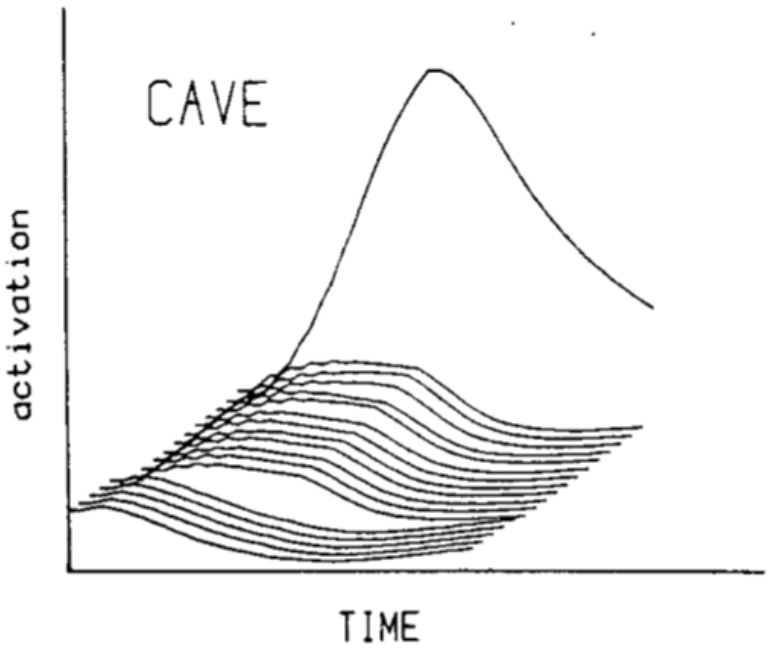


Figure 13. Activity at the word level upon presentation of *CAVE*, with weak letter-to-word inhibition.

We can also explain why seeing a word again, very shortly after just having seen it leads to faster identification in these terms.

Activation does not rise and fall instantaneously.

We might assume (as shown) that it falls off more slowly than it rises.

If the word is encountered again while activation is still high from the last exposure, the "baseline" for the activation is already higher than usual...

I’ve left out two factors:

how similar it is to other words (neighborhood effects)

whether you just saw a similar or related word
(associative/semantic/phonological priming)

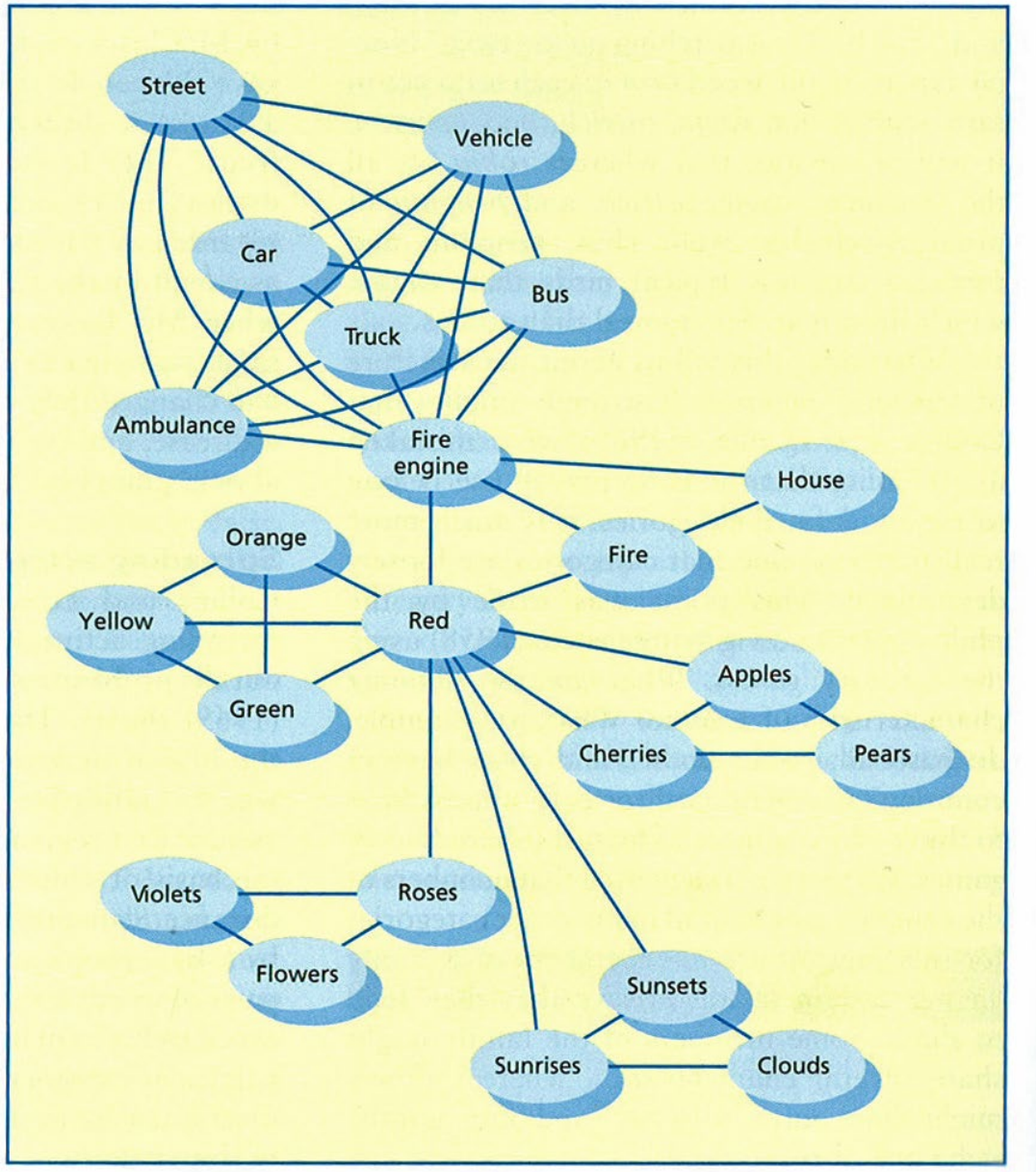
...because they require us to add something to our little network.

Spreading Activation

Remember our norms?

One way to think about how those associations are represented is to imagine a network in which the connection strengths (or distances, as here) between nodes are determined by their association.

So when you activate "lion," some of that activation travels along a **weighted connection** to "tiger."



Graphic from this lecture by Alcom and Thomas: http://www.inf.ed.ac.uk/teaching/courses/inf1-cg/lectures/12/

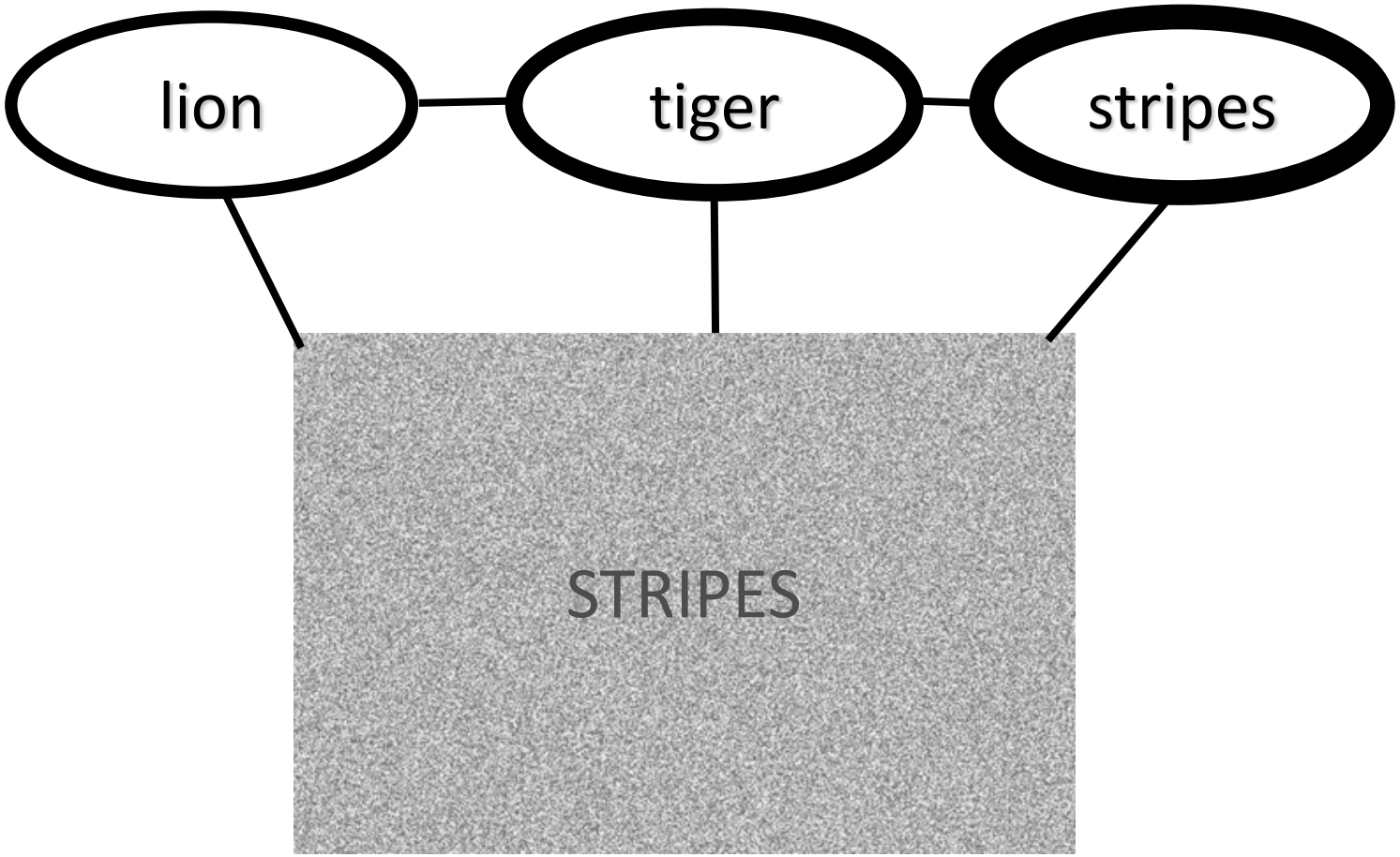
A neat experiment...

Depth of Automatic Spreading Activation: Mediated Priming Effects in Pronunciation but Not in Lexical Decision

David A. Balota
Washington University

Robert F. Lorch, Jr.
University of Kentucky

Lexical decision and pronunciation experiments were conducted to investigate whether activation automatically spreads beyond directly associated concepts within the memory network. Prime-target pairs were constructed such that there was a relation between the prime (e.g., *lion*) and the target (e.g., *stripes*) only through a mediating concept (e.g., *tiger*). The lexical decision results yielded facilitation of directly related priming conditions (e.g., *lion–tiger* and *tiger–stripes*); however, the mediated condition (e.g., *lion–stripes*) did not facilitate performance compared to either a neutral prime or an unrelated prime condition. In contrast, the pronunciation results yielded facilitation of both directly related and mediated priming conditions. The results were viewed as supporting the notion that activation spreads beyond directly related concepts in semantic memory. It is suggested that characteristics of the lexical decision task masked the appearance of a mediated priming effect. Implications of an automatic spread of activation beyond directly related concepts are discussed.



The idea is that activity flows throughout the network, so activating “lion” can prime “stripes,” even though these are not *directly* associated. Thus *mediated priming*.

- This is very representative of a lot of psycholinguistics:
  - careful study design
  - small effects
  - Counterintuitive: why would “stripes” prime “lion?”

Table 2
Mean Pronunciation Latencies as a Function of Prime Condition and Stimulus Onset Asynchrony (SOA) in Experiment 2

	Prime condition			
SOA	Related	Mediated	Neutral	Unrelated
250	545	558	570	574
500	553	558	583	576
M	549	558	577	575

- ....now we will briefly consider three more “big picture” issues

Belke et al.

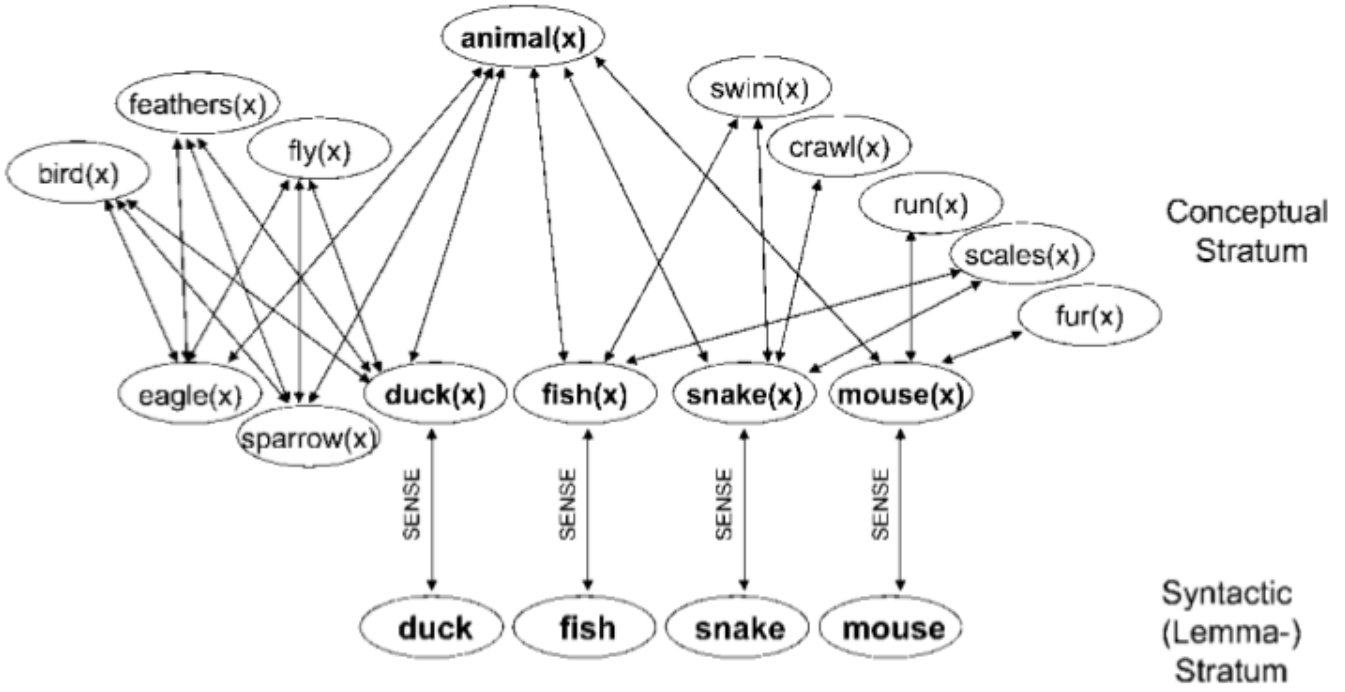


Figure 1. Fragment of the conceptual and syntactic strata, as assumed in WEAVER++ (adapted from Roelofs, 1992, and Levelt et al., 1999). Note that the arrows represent the flow of activation in the network rather than types of connection. Different types of relations exist between lexical concepts and their feature nodes, such as *is-a*, *can* and *has*. For the sake of readability, not all connections among related nodes are presented.

If we just take a naïve view of spreading activation, we might imagine that a whole block of similar pictures presented together would lead to *faster* naming times…

This model helps us understand how spreading activation can lead to *slower* naming times in some cases.

Theme 4: Whose house?



Ludwig Lichtheim

Lichtheim’s house.

Localization and language (aphasia)



In 1861, neurologist Pierre Paul Broca presented the case of "Tan" who had lost the ability to speak after a brain injury decades earlier.

He gave the presentation to the newly established Society of Anthropology, of which he was also the founding president.

Tan:

- could only produce one jargon syllable
- seemed otherwise to be "in possession of his senses"
- *in fact, seemed to understand what was being said.*

This was not the first observation of a specific loss of





Tan's brain on display at the Musee Dupuytren

Anatomical description:

- large lesion

- definitely including the third convolution of the frontal lobe (inferior frontal gyrus)

- more details on this in a moment...

Thus, we have "Broca's area:" a portion of the left, frontal lobe that is important for speech.

Consider this contemporary criticism:

- "to locate the damage which destroys speech and to locate speech are two different things" Hughlings Jackson (1874)

Localization and language (aphasia)

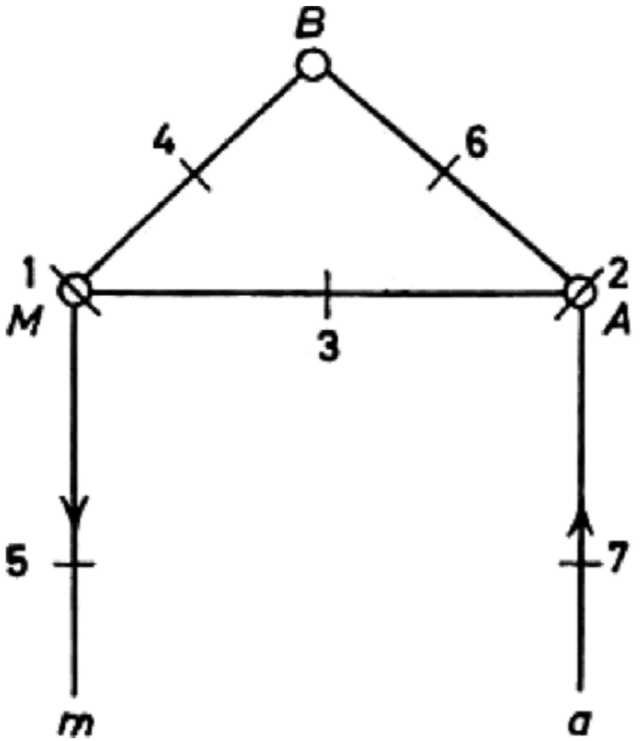


Fig. 36.5. Lichtheim's "House" model. A represents the center for sound images; M is the center for movement images; B is the center for concepts; a is the acoustic pathway and m is the motor pathway. 1–7 represent various forms of aphasia, depending on the destruction of a center or pathway.

Eling, P., & Whitaker, H. (2009). History of aphasia: from brain to language. In *Handbook of clinical neurology* (Vol. 95, pp. 571-582).

In the 1870's a number of German neurologists — Lichtheim, Wernicke — began making more systematic observations of aphasia.

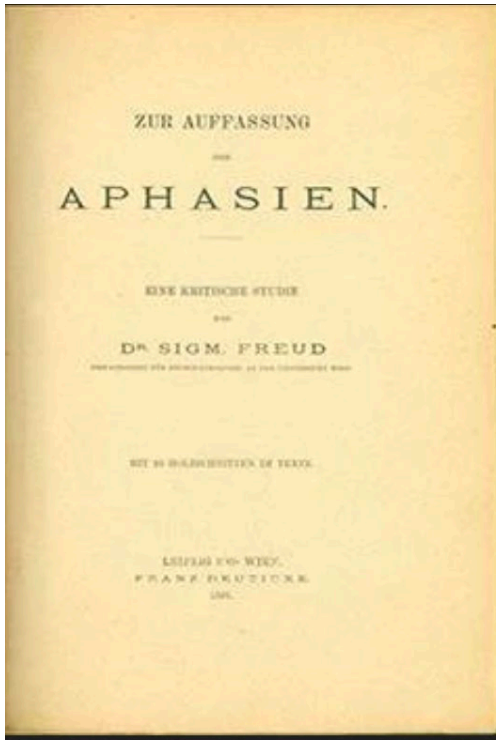
They ascribed specific losses of function to "disconnections" between "centers."

So, a loss of the ability to understand spoken words might be due to a disconnection between "A" and "B."

A loss of the ability to do verbatim repetition may be due to a disconnection between "A" and "M," and so on.

This approach of describing "centers" for specific functions is an ancestor of a topic we'll be dealing with later on in the course: "modularity"

A holist critique


ZUR AUFFASSUNG

APHASIEN.

EINE KRITISCHE STUDIE

Dr SIGM. FREUD

Sigmund Freud (!) critiqued the “diagram-drawing” school of aphasia research in his early neurological research.

He felt it was naïve to expect the brain to be organized according to psychological categories like “word forms” or “concepts.”

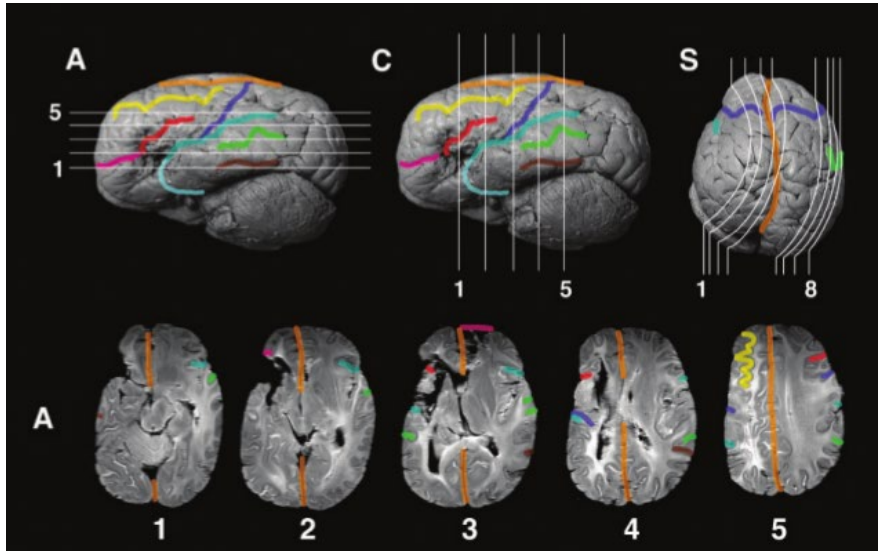
He proposed instead that language emerged from “fields” connecting perceptual and motor abilities to “drives.”

This is a durable tension in cognitive neuroscience:

*Can we directly relate **brain area** functions and **cognitive** functions or not?*

Is the brain really organized according to our descriptions of cognitive phenomena?

Back to Leborgne’s brain…



Using MRI, Dronkers and colleagues were able to assess the extent and location of damage in Broca's landmark cases very precisely.

The lesion is very deep, and includes the insula.

There is also a lot of white matter damage to the superior longitudinal fasciculus (which, basically connects the front and back of the cerebral cortex).

Indeed, other work by this group suggests that the insula, not Broca's area, is strongly associated with *speech apraxia* (the condition Broca described in his patients).

Occasional Paper:
N. F. Dronkers, O. Plaisant, M. T. Iba-Zizen, and E. A. Cabanis
Paul Broca's historic cases: high resolution MR imaging of the brains of Leborgne and Lelong
Brain (2007) 130 (5): 1432-1441 first published online April 2, 2007 doi:10.1093/brain/awm042

Theme 5: The functional and positional level of the “consensus model.”

The Functional Level -- Lemmas

We can't just say that we select the "words" that we're going to produce at the functional level…

Why not?

Consider this error:

Where'd I put it when I brought it in → Where'd I bring it when I put it in.

So, the verbs switched places, but the actual words that were produced reflect a change in tense that must have been applied later.

This couldn't have happened if the *word* "brought" was selected. Rather, we say the lemma BRING was selected, and was expressed as present tense…



No, not Lemmings.

(The term "lemma" is borrowed from mathematics, where it refers to a **proposition** in an **argument**.)

And they *do not* just follow each other blindly off cliffs, BTW.

The Functional Level

Why do people so often say the opposite of what they mean?

(30) (a) I really like to – hate to get up in the morning
(b) It's at the bottom – I mean – top of the stack of books
(c) This room is too damn hot – cold
(d) the oral – written part of the exam

Antonyms are actually quite *similar* at a number of levels:

- semantically related
- strongly associated

… it's like a perfect storm in the spreading activation framework…

further:

- can take on the same semantic roles
- same word class (can take on same syntactic roles)

...so the rest of the speech production system can handle them…

So it's easy for activation to spread from a lemma to its antonym.

The Functional Level

Functional (Semantic) roles.

Remember:

{The plums} {in the icebox} were eaten {by me}
{Undergoer} + {Location} + Verb + {Agent}

Alternatively:

I + have eaten {the plums} {in the icebox}.
Agent + Verb + Undergoer + Location

So, we have a simple *event structure*, and we can start slotting lemmas into semantic roles. We know that the Verb will be TO EAT, the Agent of the verb will be ME, and that PLUM will be the Undergoer, etc.

Note there are syntactic consequences for arranging these roles in different orders, with consequences for how we turn the lemma for ME into a word.

That will happen at the *Positional Level*, in our next session…

At the functional level:

We have decided on a message to convey.

We have selected appropriate lemmas.

We have associated the lemmas with semantic roles.

…this is quite a bit, and yet we're nowhere near putting a clause together, much less saying it out loud.

The Positional Level

Let's take the simple-seeming sentence we used as evidence that lemmas are a useful concept:

Where'd I bring it when I put it in?

By the time we're done at the functional level, we know:

1. that we have two Verbs PUT and BRING
2. that I/ME is the Agent of both verbs
3. that IT is the Undergoer of both verbs

What else?
What is the temporal relationship between the verbs?

Expressing all of those things syntactically is what goes on at the *Positional Level*.

If we skip the part where we organize the sentence into a *question*, and work out the appropriate tenses for the verbs, etc., we get something like:

Where'd I [verb, root form] it when I [verb, past] it in?

(Actually, rather a lot of syntactic work had to happen to get that first clause right, because we have both Wh- movement and an auxiliary verb to form the question.)

What we are paying attention to here is that that structure of the first clause requires the base form of the lemma in English.

…and then the next clause requires the past form.

So, what we have at the Positional Level is a syntactic structure with slots for all of the lemmas we activated at the Functional Level, along with all of the other particles required by the grammar.

An error like "Where'd I bring it when I put it in," can be thought of as arising while the slots generated by the Positional Level are being filled with lemmas activated at the Functional level.

A similar error:

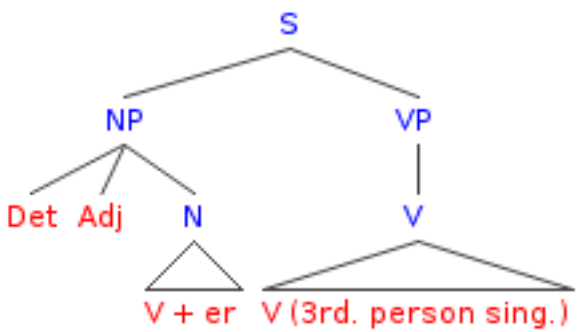
a language needer learns

How is this similar to the other sentence?

two lemmas got switched when they were being fed into the positional level
the positional level stayed basically the same

How is it different?

look at the Noun
it's actually a Verb plus an affix!
*this is **derivational morphology...***



Another similar error:

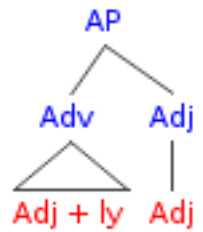
He did that easy enoughly

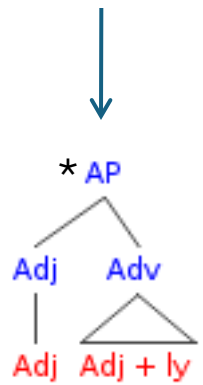
How is this similar to “a language needer learns”?

both have a “stranded affix”
you can make an adjective into an adverb with +ly

How is it different?

in this case, something also happened when setting up the positional level.



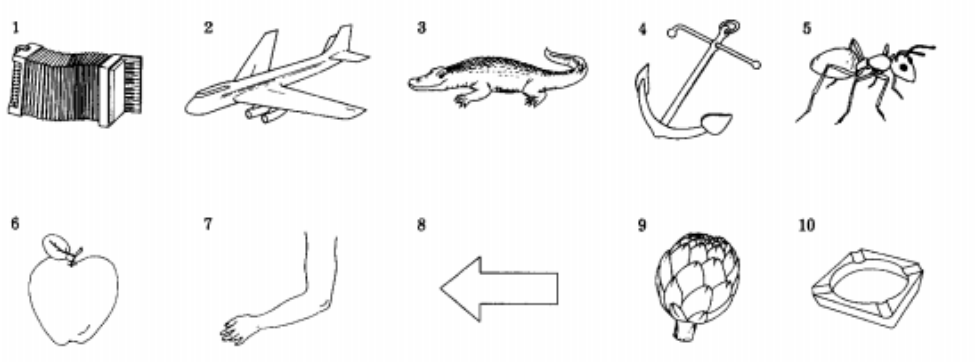


Theme 6: Stimulus control

A classic cautionary tale:
The frequency effect, and its confound with length

Oldfield and Wingfield (1965) asked people to name pictures of objects.

They found a log-linear relationship between frequency and response time.



Note that frequency is related to word length, however, even in this tiny sample.

Let's look at this quantitatively, using a web-based tool...

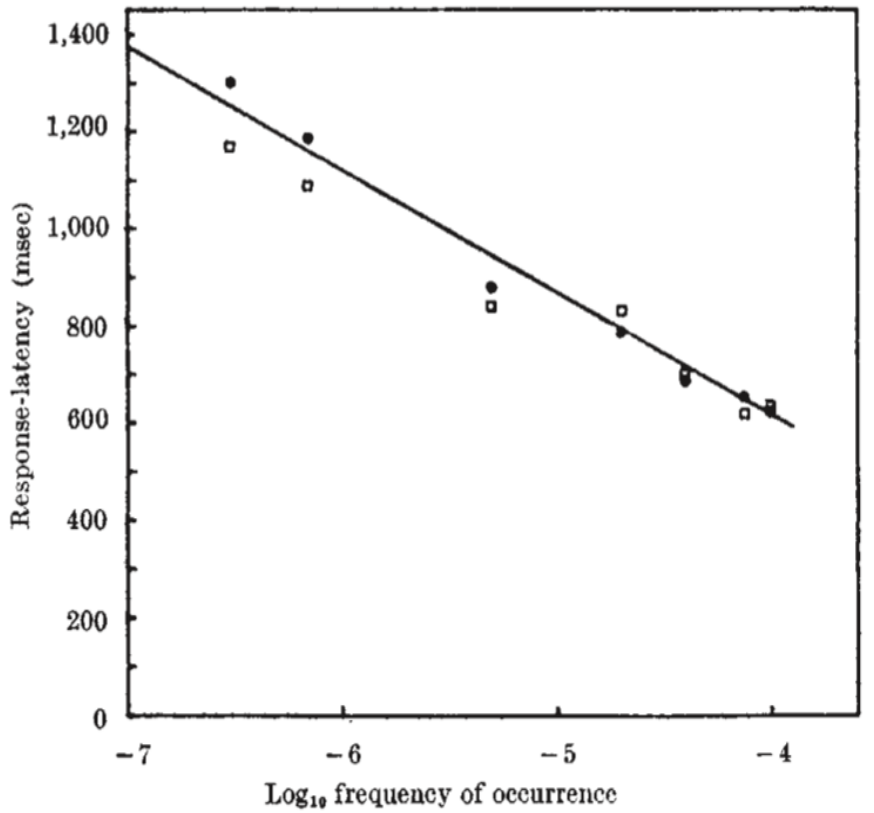


Fig. 1. Response-latency in naming objects versus $\log_{10}$ frequency of occurrence of the name in English. ●, Picture set 1; □, picture set 2

Oldfield, R. C. & Wingfield, A. (1964) The time it takes to name and object. *Nature, 202,* 1031-1032

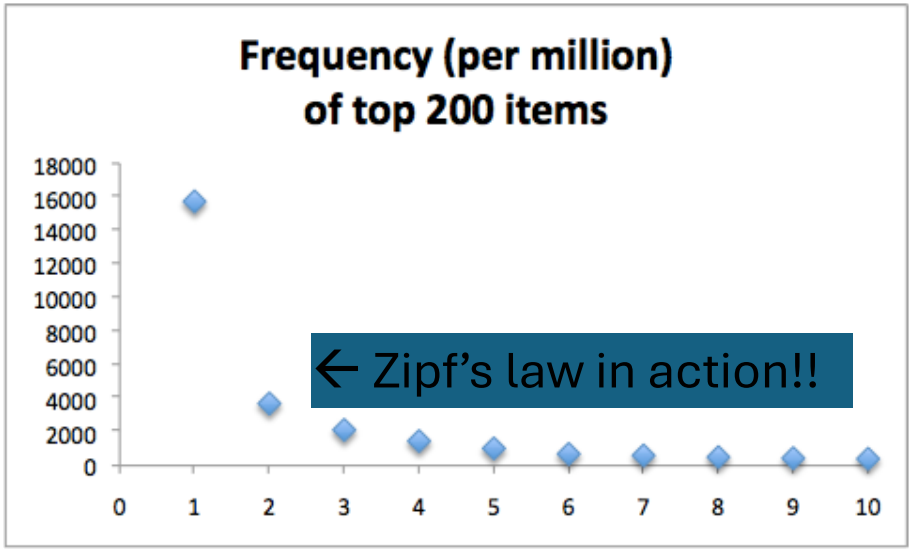
...these line drawings are from Snodgrass & Vanderwart (1980)...

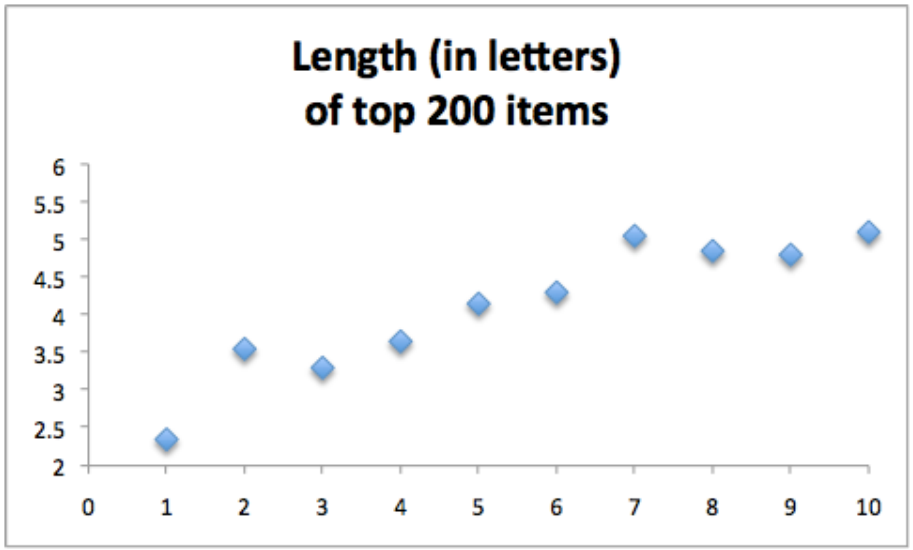
The 20 most frequent words in English

Word	Length	Freq_KF
the	3	69,971
of	2	36,411
and	3	28,852
to	2	26,149
a	1	23,237
in	2	21,341
that	4	10,595
is	2	10,099
was	3	9,816
he	2	9,543
for	3	9,489
it	2	8,756
with	4	7,289
as	2	7,250
his	3	6,997
on	2	6,742
be	2	6,377
at	2	5,378
by	2	5,305

20 more high-frequency words

century	7	207
Washington	10	206
usually	7	206
north	5	206
therefore	9	205
plan	4	205
leave	5	205
top	3	204
sound	5	204
million	7	204
evidence	8	204
black	5	203
strong	6	202
hard	4	202
various	7	201
value	5	200
type	4	200
surface	7	200
says	4	200
play	4	200



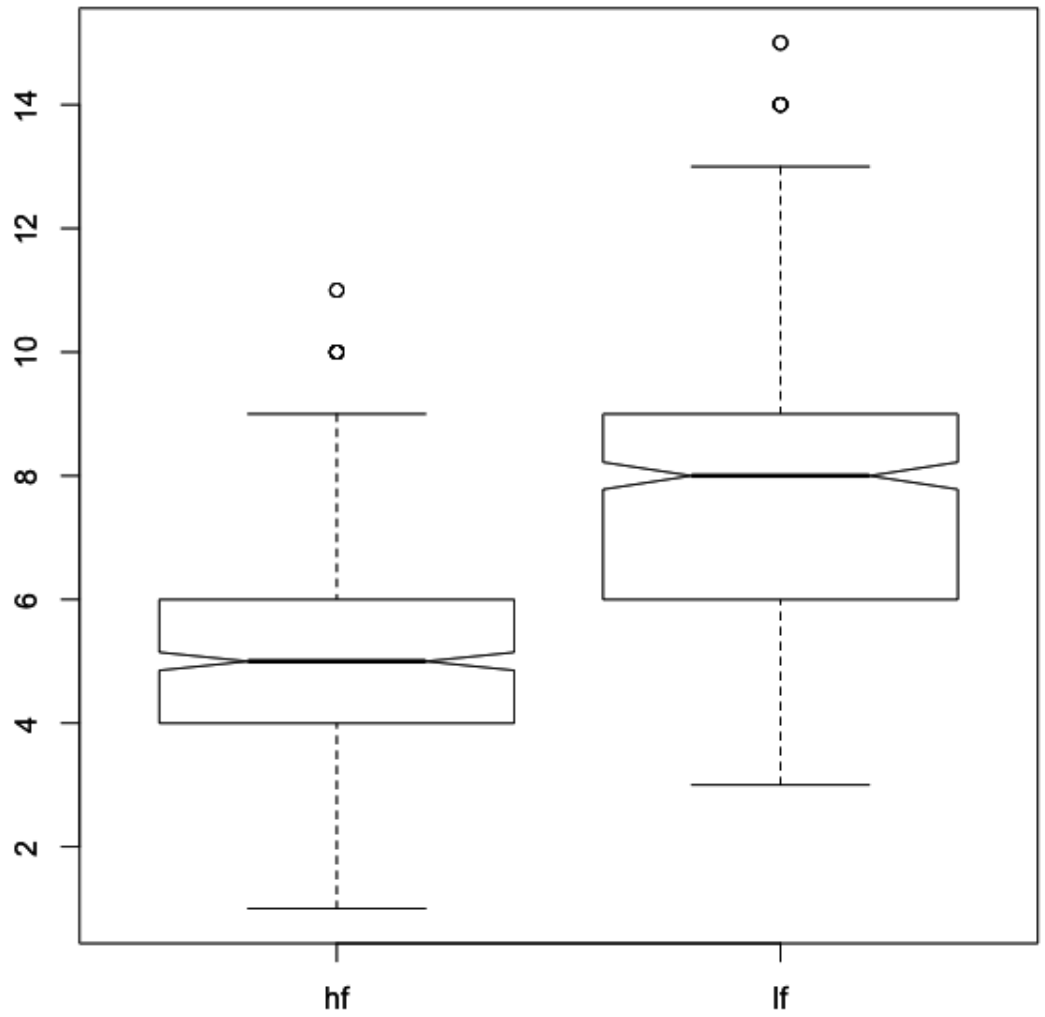


What I did here was take the 200 most frequent words in an old corpus, and put them into ten "bins" of twenty words each.

Notice:

1. The mean frequency drops off very dramatically from the most frequent items to the next most frequent, etc.

2. Length is inversely related to frequency.



Problem: HF words are just generally shorter than LF words, so this will always happen if you select LF words at random.

Solution: *don't select LF words at random!*

If you select long-ish HF words and short-ish LF words, you can match on length while manipulating frequency…

Oldfield and Wingfield eventually addressed this by matching their high- and low-frequency words for length.

They found a robust frequency effect, so they concluded that frequency effects are not merely length effects in disguise.

Really rather a lot of psycholinguistics can be characterized as an attempt to demonstrate that some observed effect is not just an effect of some other confounding variable…

Theme 8: Cyclic semantic blocking

THE QUARTERLY JOURNAL OF EXPERIMENTAL PSYCHOLOGY
2005, 58A (4), 667–692

Psychology Press
Taylor & Francis Group

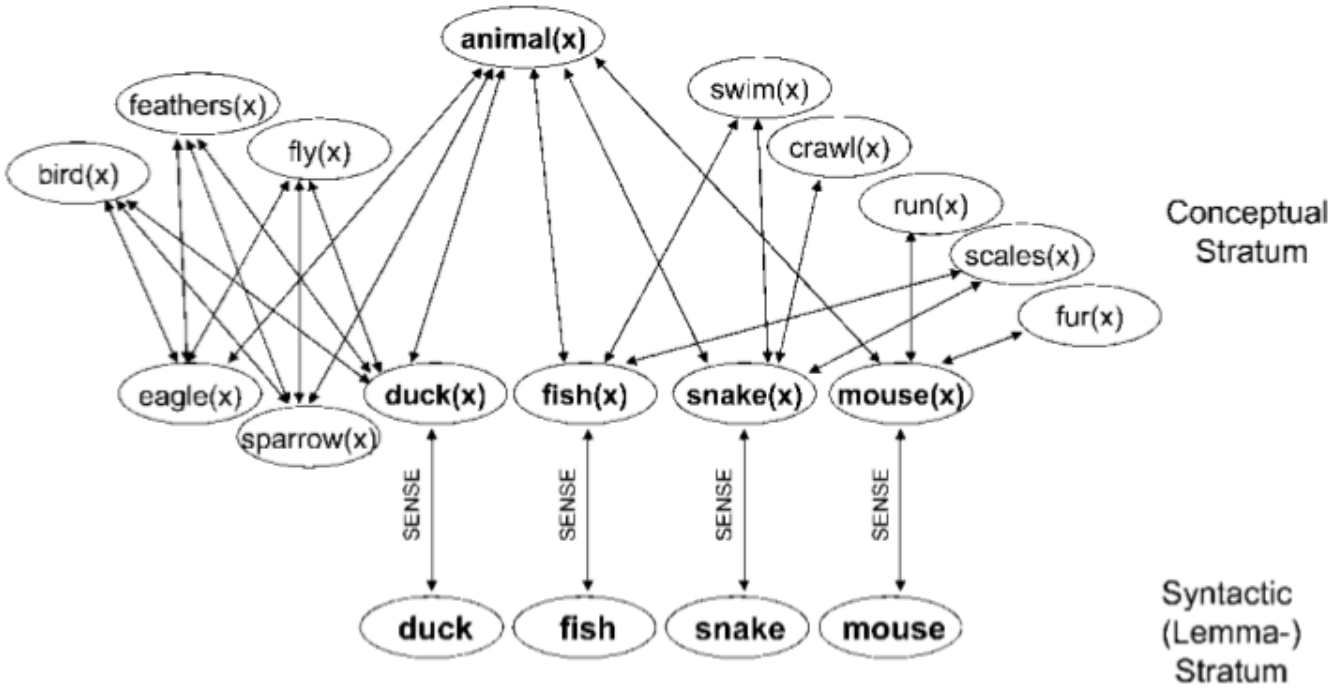
Refractory effects in picture naming as assessed in a semantic blocking paradigm

Eva Belke and Antje S. Meyer
The University of Birmingham, Birmingham, UK

Markus F. Damian
University of Bristol, Bristol, UK

In the cyclic semantic blocking paradigm participants repeatedly name sets of objects with semantically related names (*homogeneous sets*) or unrelated names (*heterogeneous sets*). The naming latencies are typically longer in related than in unrelated sets. In Experiment 1 we replicated this semantic blocking effect and demonstrated that the effect only arose after all objects of a set had been shown and named once. In Experiment 2, the objects of a set were presented simultaneously (instead of on successive trials). Evidence for semantic blocking was found in the naming latencies and in the gaze durations for the objects, which were longer in homogeneous than in heterogeneous sets. For the gaze-to-speech lag between the offset of gaze on an object and the onset of the articulation of its name, a repetition priming effect was obtained but no blocking effect. Experiment 3 showed that the blocking effect for speech onset latencies generalized to new, previously unnamed lexical items. We propose that the blocking effect is due to refractory behaviour in the semantic system.

Belke et al.



In speech production, we have to distinguish carefully between *activation* and *selection*.

Spreading activation from, e.g., *duck* to *fish* means that activating one can help activate the other. But this can cause problems when our task is to *select* the correct lemma from a bunch of active ones…

The WEAVER++ model assumes that selection time is determined by a ratio of the activity of the *most active* lemma and the *overall activation of all lemmas*.

Thus, when there's feedback from the category level, this slows down lemma selection because of all the collateral activation for items from the same category.

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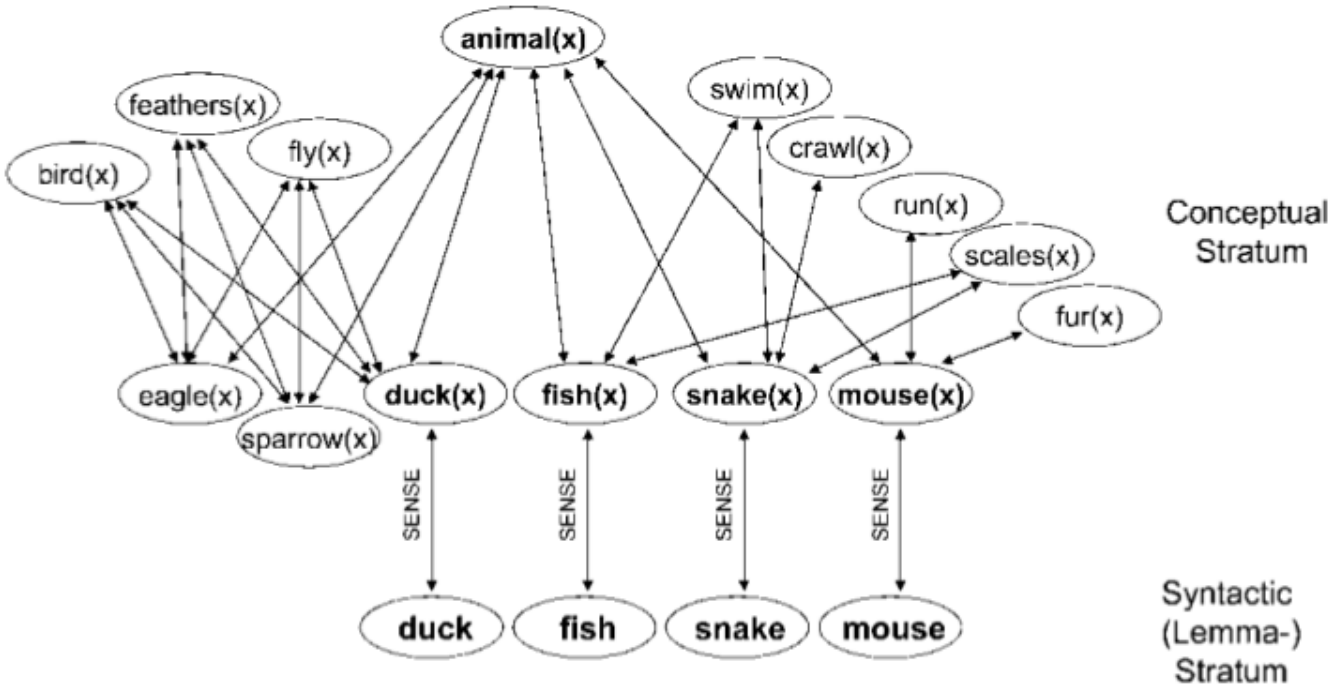
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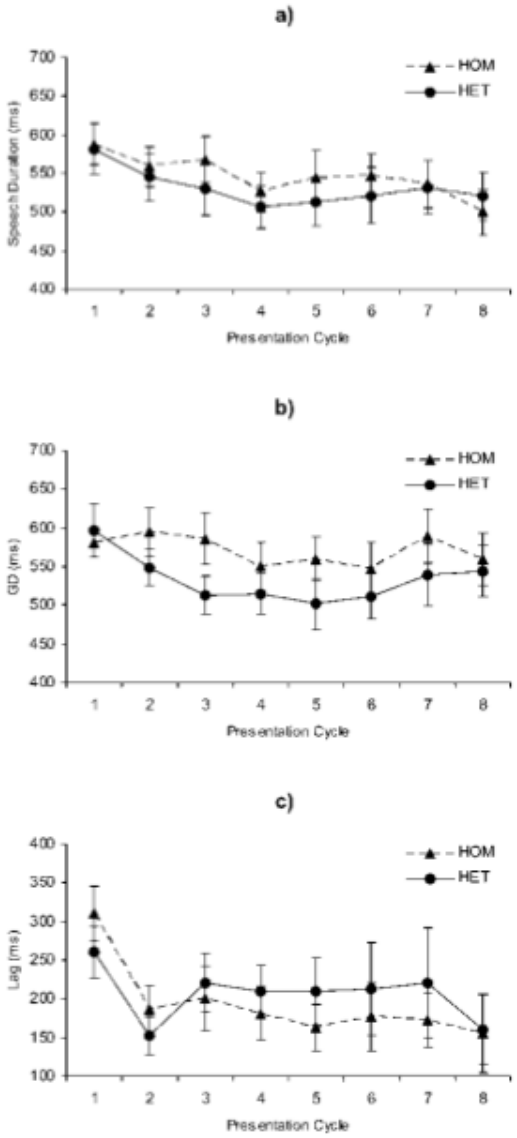


Figure 4. Experiment 2: (a) Mean speech durations, (b) gaze durations (GD), and (c) eye–speech lags by context and presentation cycle. The error bars represent one standard error.

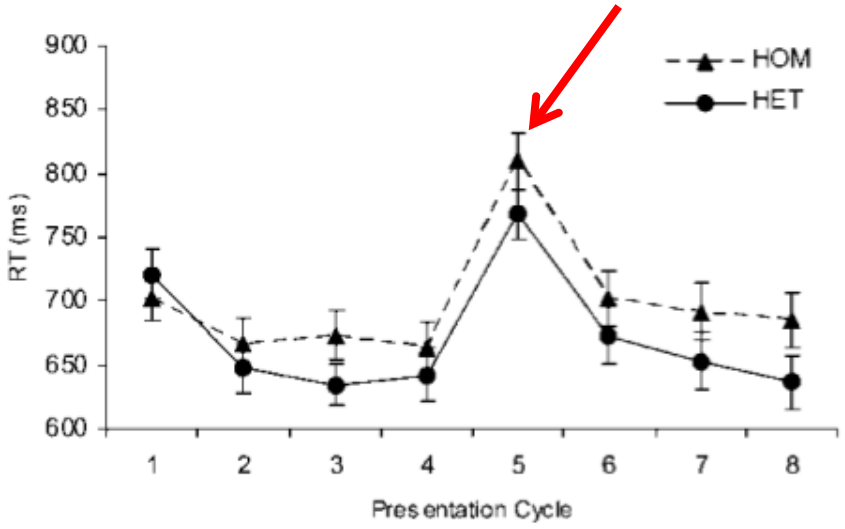
Overall duration of speech decreases over the course of the experiment, but doesn't show a blocking effect.
→*gesture level*

Gaze duration shows a blocking effect, but only after the first cycle.
→*lemma selection*

There's no effect on lag. Numerically, the blocking effect seems to go in the opposite direction, but this is not significant.
→*phonological encoding*

Experiment 3
Does blocking generalize to new items in the same category?

New items introduced



Note that the repetition priming effect goes away in Cycle 5…

Overall, performances slows down dramatically, but the blocking effect is still there.

Compare the effect on new items from a blocked category to the items in Cycle 1…

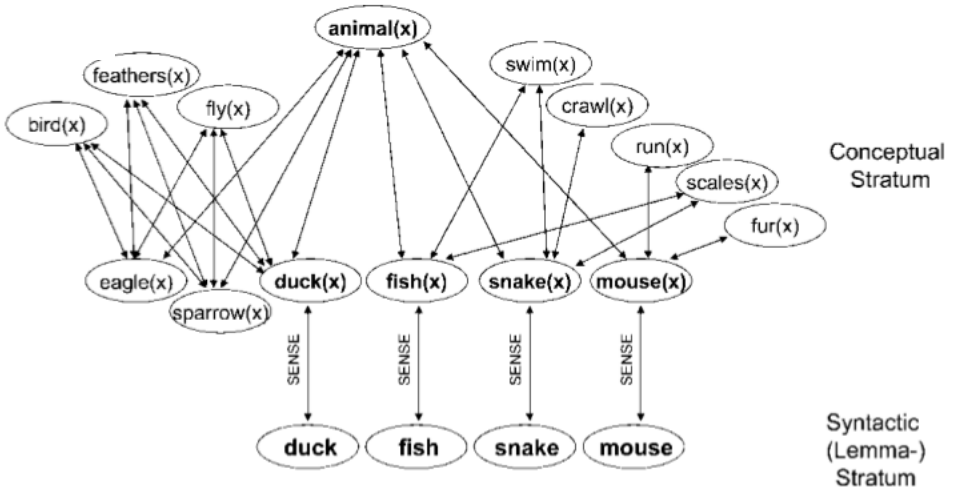


Figure 1. Fragment of the conceptual and syntactic strata, as assumed in WEAVER++ (adapted from Roelofs, 1992, and Levelt et al., 1999). Note that the arrows represent the flow of activation in the network rather than types of connection. Different types of relations exist between lexical concepts and their feature nodes, such as *is-a*, *can* and *has*. For the sake of readability, not all connections among related nodes are presented.

How do Belke et al. isolate the blocking effect in the model:

- not visual recognition
    - normed/matched stimuli
    - control experiment

- not phonological encoding (?)
    - words are not phonologically similar
    - gaze duration, not eye-speech lag is impacted

- not just a specific effect of repeating lists
    - generalizes to new items